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# ELIAIR

## WHEN VISION FAILS: PREDICTING AIR QUALITY AND LANDING SAFETY VIA LSTM-XGBOOST FUSION

## INTRODUCTION

This project aims to predict the AQI(Air Quality Index) in Sarajevo based on previous data, and classify the safety of a plane landing on the Sarajevo International Airport. The repository contains all necessary datasets, as well as code for training our models. The innovation in our project comes from the fact that we utilized two approaches for the two problems- LSTM for predictions, and XGBoost for the classification. Overall, the project focuses on one of the most important issues in Sarajevo, which is air pollution and its effect on plane-landing safety. Our goal was to build a model which can successfully predict, classify and help with the stated issue.

## METHODOLOGY

We used two separate models in sequence. The first component- LSTM, is a two-layer network which was trained on hourly data spanning March 2022 to March 2026. It takes a 24-hour sliding window of 14 relevant input features- European AQI, PM2.5, PM10, etc. Then, it predicts the next hour's AQI. MinMaxScaler was used on features in the training set, Adam optimizer and MSE for the loss function, gradient clipping and early stopping in training. The XGBoost part is a binary classifier which classifies whether a plane can/will land or not. The data used combined flight records with AQI data, with categorical features being one hot encoded. The model runs 300 boosting rounds with max depth of 4 and a learning rate of 0.05.

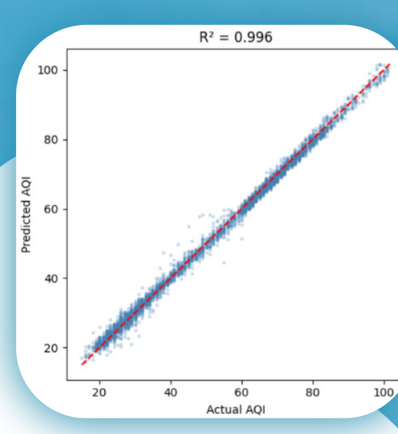
## DATASETS

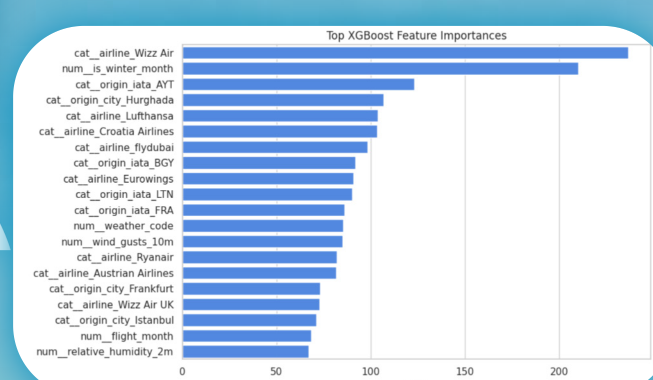
Data from different sources, such as Openaq, Open-Meteo and Flightera, was used in order to create our very own datasets- one for predicting the AQI, and another for flight-landing classification. Both datasets are provided in our repository.

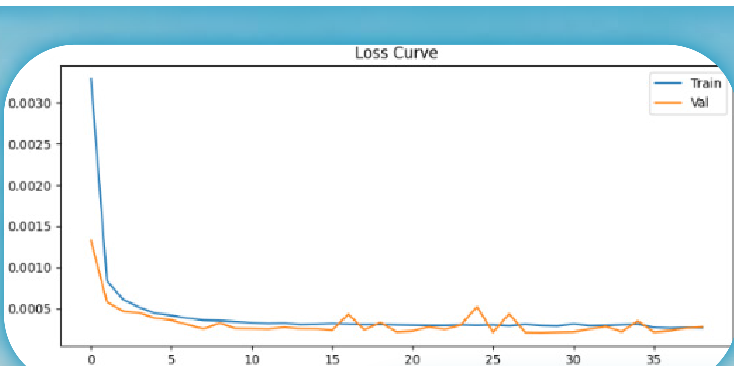
## RESULTS

The LSTM model demonstrated a strong predictive performance on the test set. An MAE of 0.957, RMSE of 1.279 and $R^2$ of 0.9964 were achieved, showing that the model captured the majority of variance in AQI values.

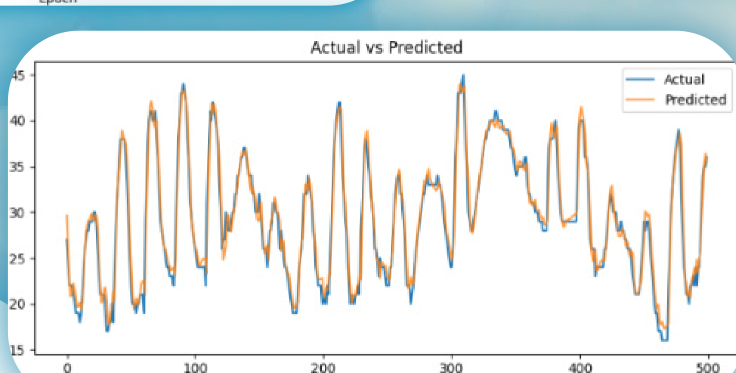
The XGBoost classifier achieved an accuracy of 92.15% and a ROC AUC of 0.7857. The performance was, still, asymmetric over classes. The model performed well on the "landed" class, but struggled on the smaller "did not land" class. Feature analysis revealed that the strongest factors for plane-landing were airline identity and winter months, which suggests that the model's landing predictions are driven more by route and seasonal patterns than by air quality conditions.








## FUTURE STEPS

Feature engineering- right now, the two models are trained independently. A future step of ours is definitely connecting the two so that the AQI prediction becomes an input feature of the XGBoost classifier.

Improving the classifier for more accurate and precise predictions by addressing class imbalance and refining the labeling of negative outcomes.

Expanding the model for all cities in Bosnia and Herzegovina with airports- Tuzla, Banja Luka, Mostar.

Getting previous landing data directly from airports, not just public sites, and exploring what happens with the planes who fail to land on the airport they were supposed to.

## CONCLUSION

ELIAIR shows that air quality forecasting and flight safety assessment can be meaningfully connected through a hybrid machine learning pipeline. The LSTM component successfully captured the dynamics of Sarajevo's pollution patterns and produced highly-accurate hour-ahead AQI predictions. The XGBoost classifier then translated environmental and operational context into landing safety assessments, demonstrating that the meteorological conditions, seasonality and route characteristics, all combined, carry predictive signals for flight outcomes. The two models currently operate in parallel, which will be changed into operating in sequence in the future. The classifier's difficulties with unsafe landing events also highlights the challenge of working with real-world aviation data.

Overall, ELIAIR lays a solid foundation for AI-assisted environmental risk assessment in aviation, with its design making it a promising basis for further development toward a deployable decision-support tool for airports.